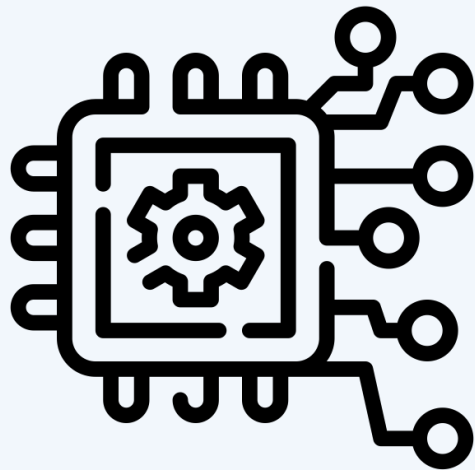
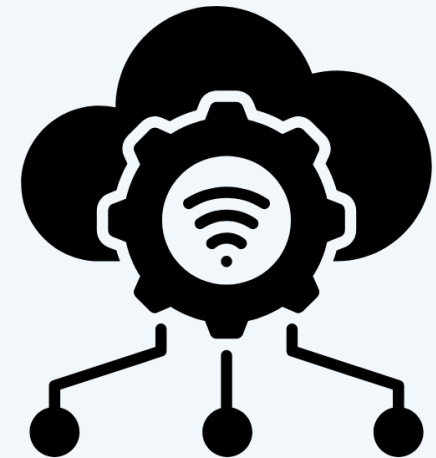


Internet of Things



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ICT, PDEU



Course Overview

- Unit 1 Introduction to IoT
- Unit 2 Technology for IoT
- Unit 3 Security for IoT
- Unit 4 IoT Applications

Unit 1: Introduction to IoT (5 Classes)

- Introduction to IoT
- Technology evolution
- IoT Architectures
- Resource management
- Communication protocols
- Standards and regulations

Unit 2: Technology for IoT (7 Classes)

- Programming for IoT
- Different frameworks for IoT
- Virtualization concepts
- Embedded platforms
- Cloud based deployment
- Distributed data analysis

Unit 3: Security for IoT (8 Classes)

- Security issues in the Internet of Things
- Robustness and Reliability
- High risk issues, Approaches and New Paradigms
- Authentication for devices
- Different methods of securing the Internet of Things (IoT).

Unit 4: IoT Applications (6 Classes)

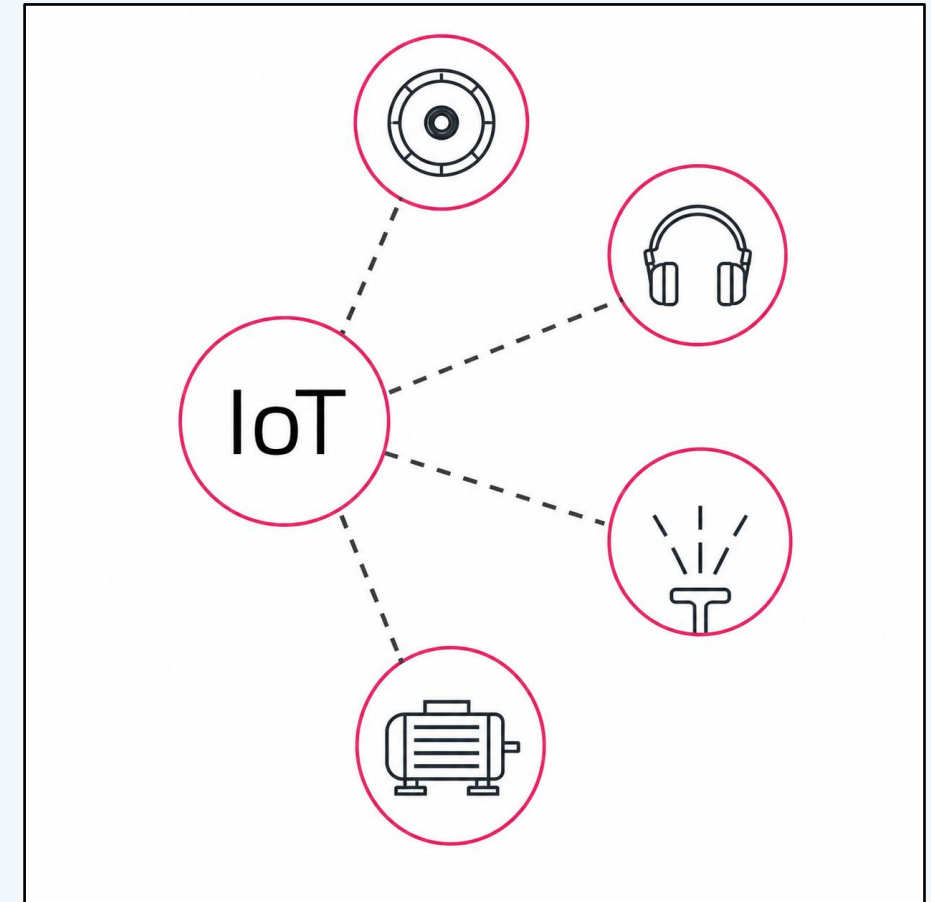
- Applied Internet of Things
- Vehicle technology, smart cities, smart agriculture
- IoT based industrial processes
- Cloud-Based Smart supply chain Management
- Case Studies Illustrating IoT Design
- Data Analytics for IoT
- Data Aggregation for the IoT

Unit 1 - Introduction to IoT



Introduction

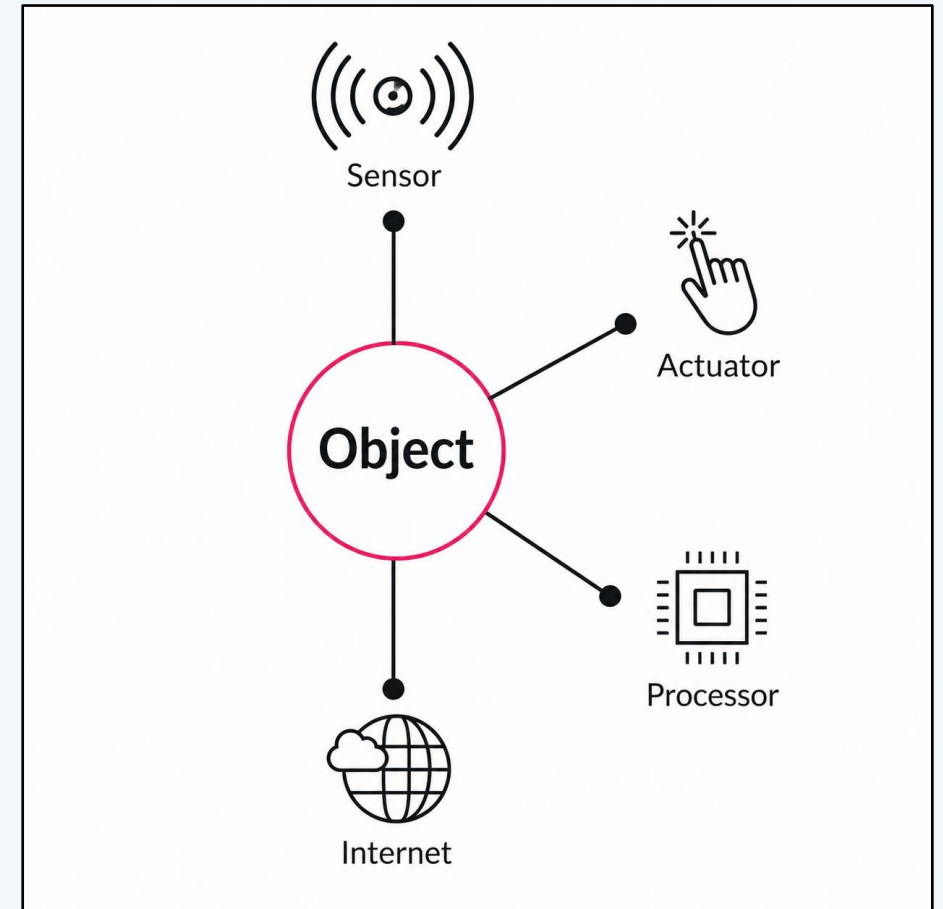
- Primary goal of IoT technology is day today devices
- IoT is configuration, controlling and networking of everyday devices



Introduction

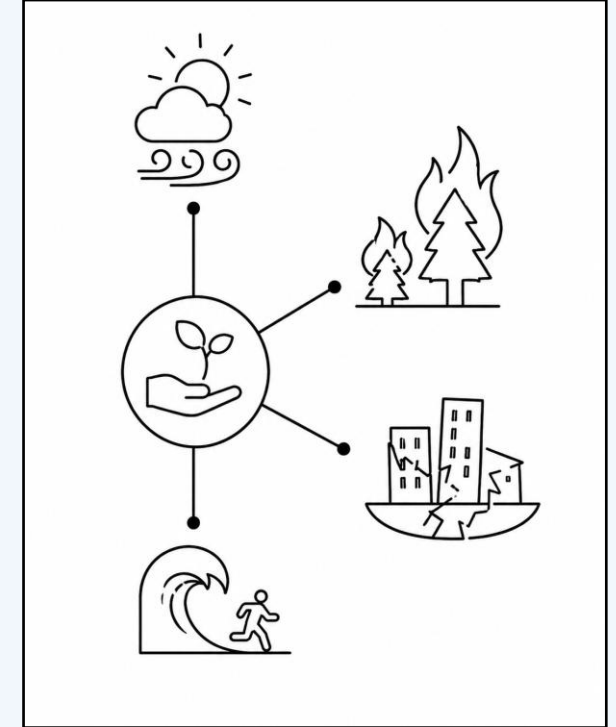
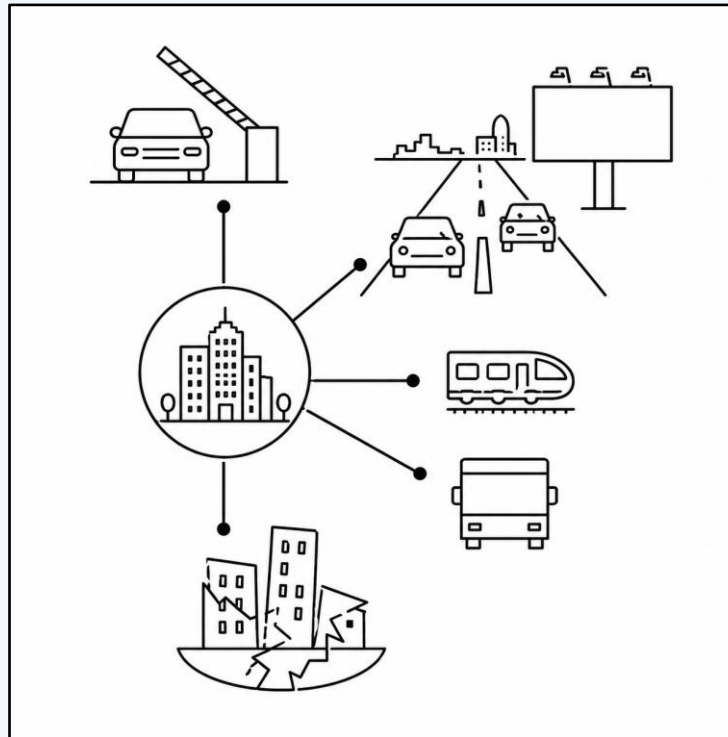
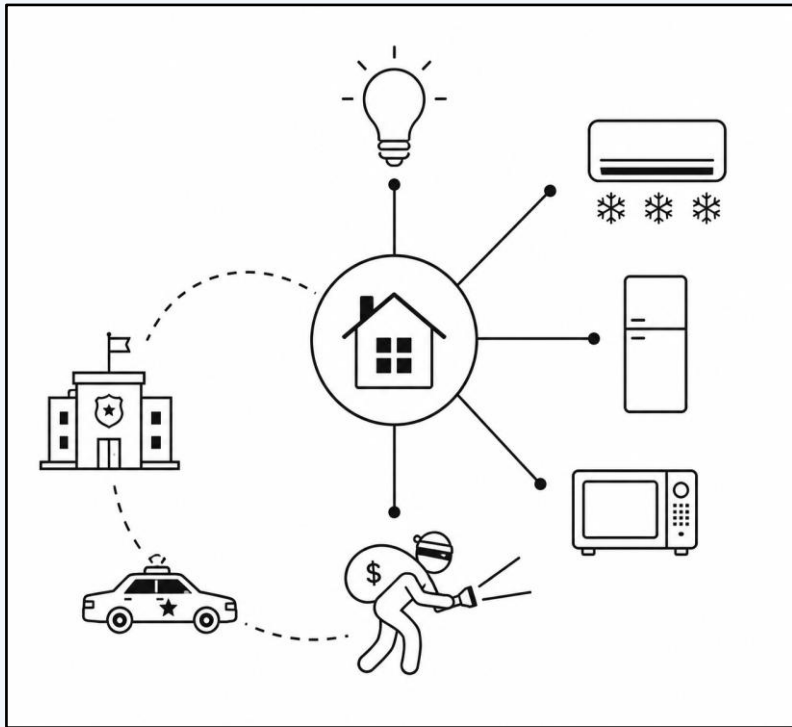
Basic building blocks of IoT device

- Physical object
- Sensor
- Actuator
- Processor
- Internet



Introduction

IoT Applications



Internet of Things (IoT)

- Network of interrelated devices that connect and exchange data with other IoT devices and the cloud
- The Internet of Things (IoT) describes the **network** of physical objects—“**things**”—that are embedded with **sensors, software, and other technologies** for the purpose of **connecting and exchanging data** with other devices and systems **over the internet**

IoT Characteristics



Smart Devices



Zero /Self
Configuration



Unique Identity



Semantic
Interoperability



Common
Communication
Network



Connected
Devices



Collective
Intelligence

IoT Components - Overview

IoT Device

Resources

Controller
Service

Database

Analysis

Application
Services

Web Service

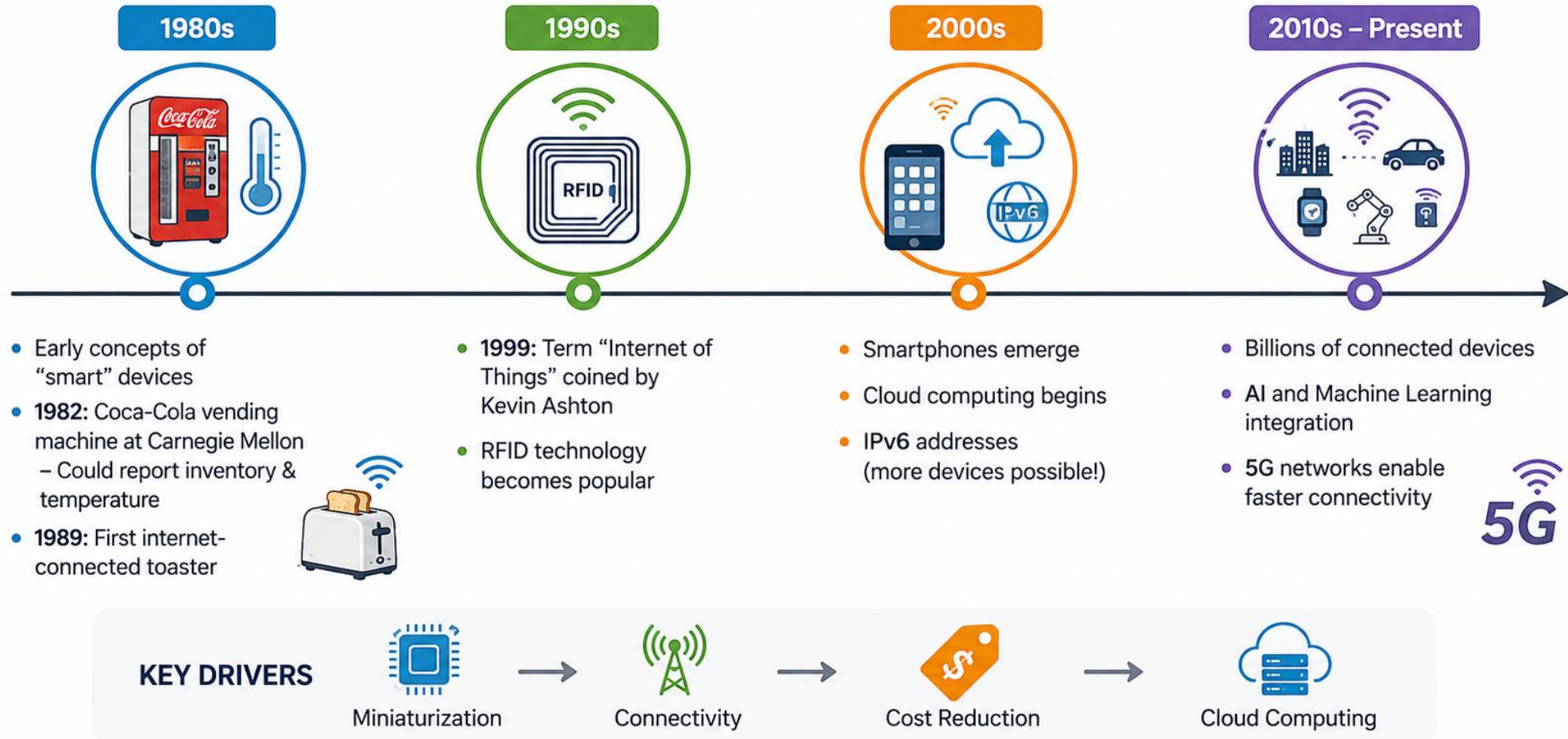
IoT Levels – Placement of IoT Components (Local/Cloud) - 6

Complexity of Application
(problem statement, requirement)



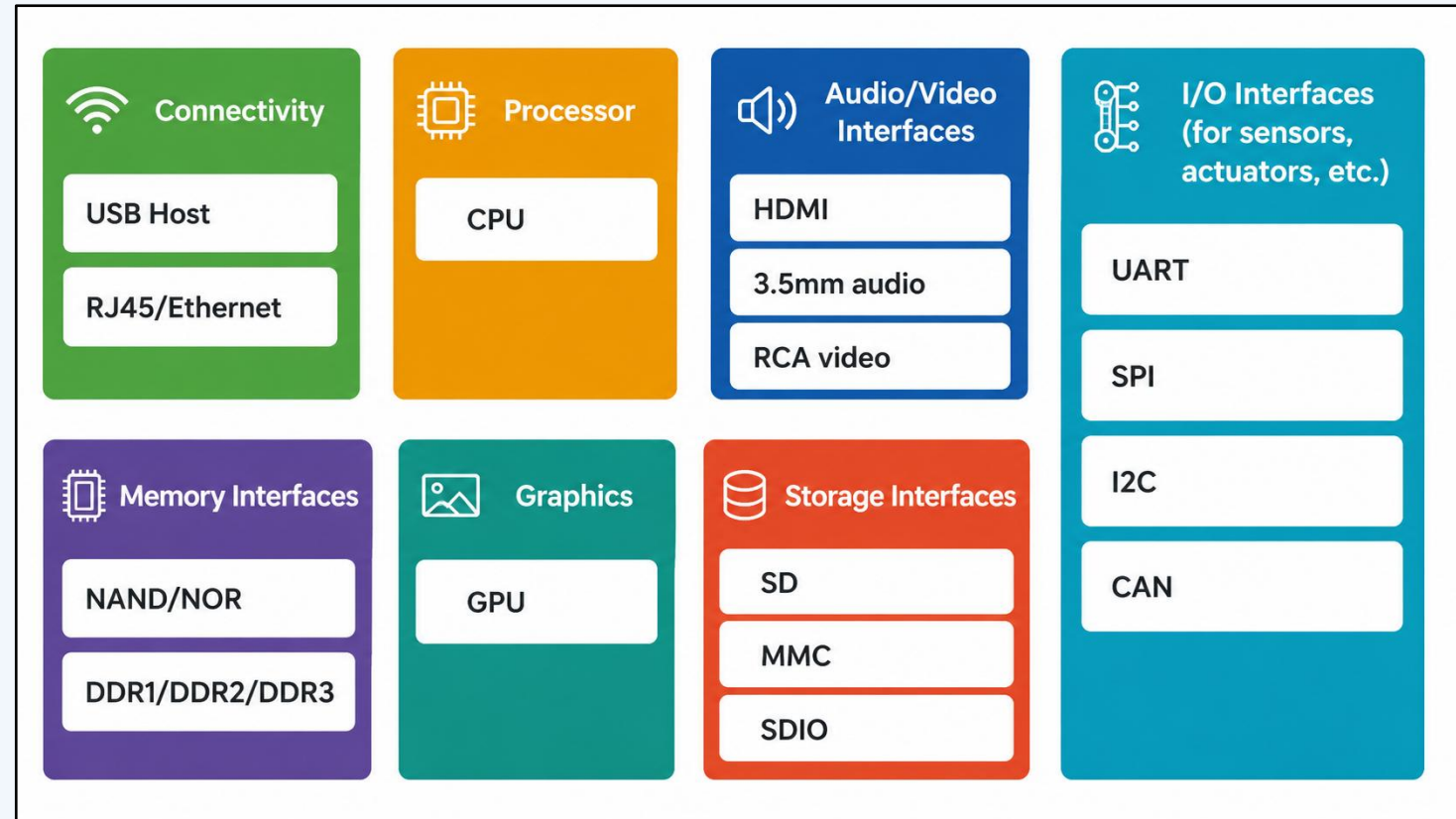
Complexity of Elements
(IoT system)

Technology Evolution



Generic IoT Device

- I/O interfaces for sensors
- Interfaces for Internet connectivity
- Memory and storage interface
- Audio/video interfaces



Things in IoT

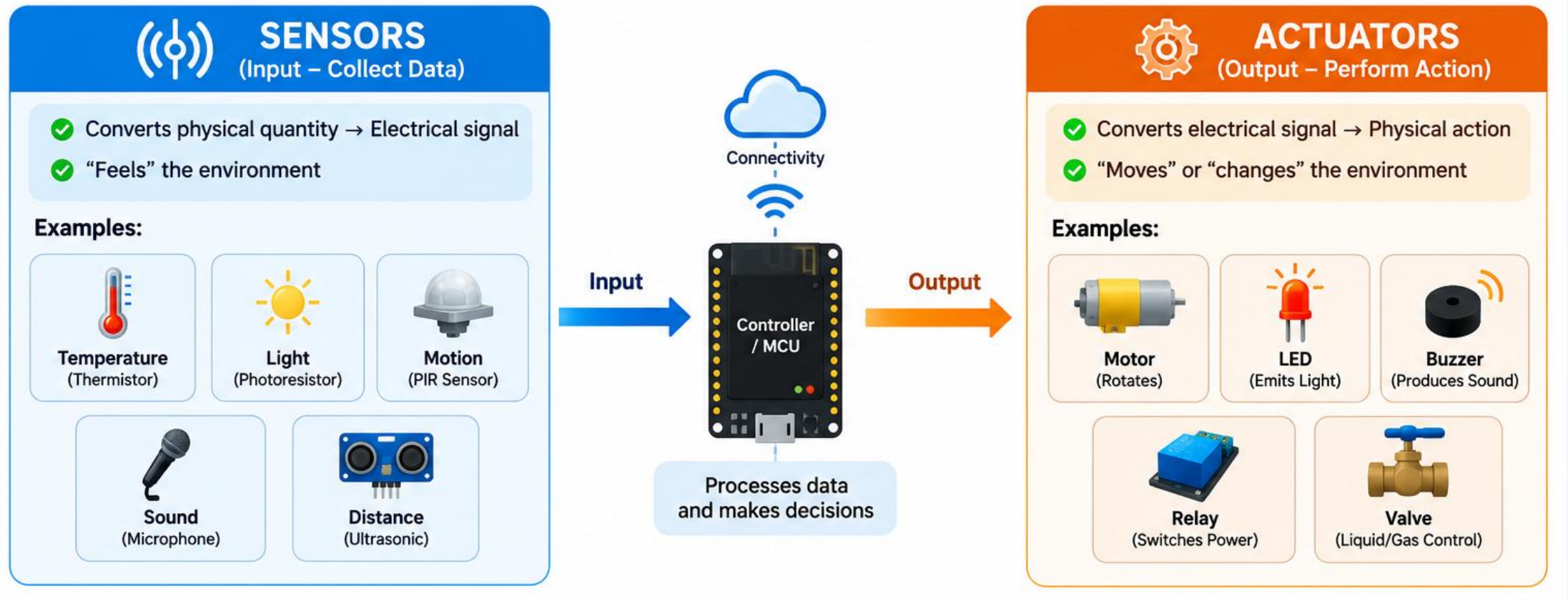


SENSORS

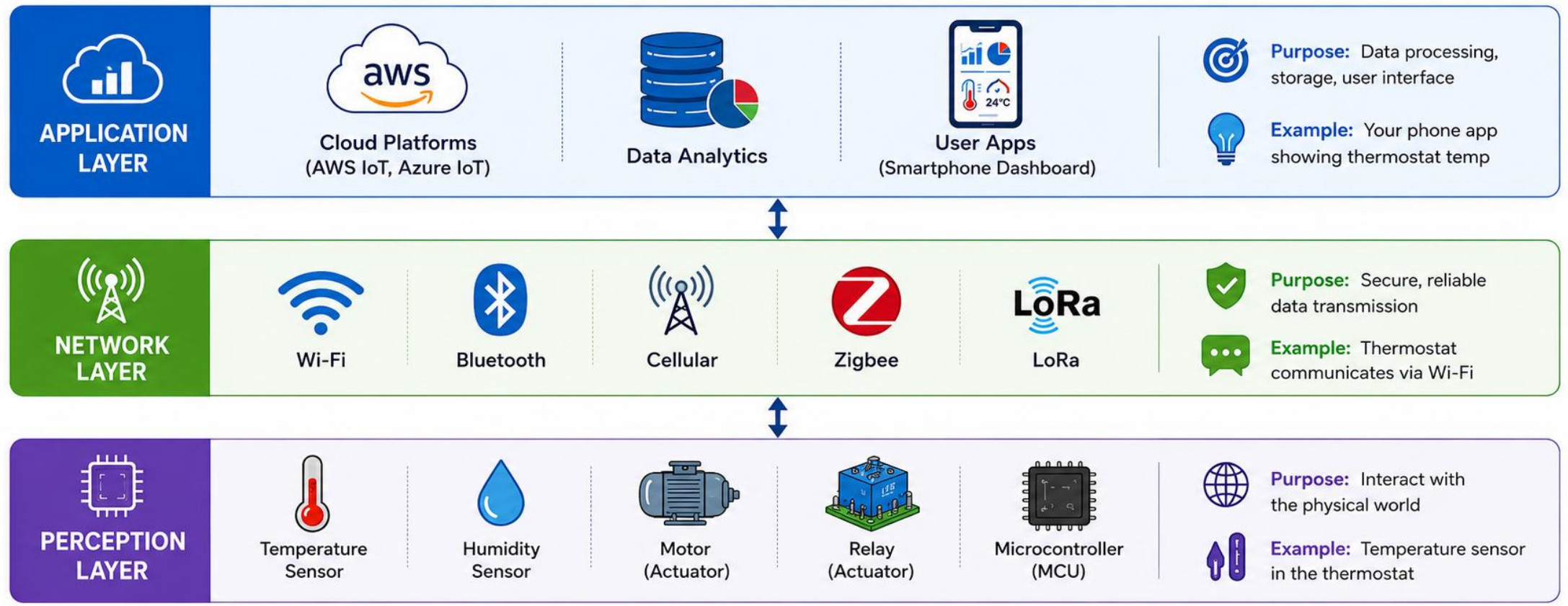


ACTUATORS

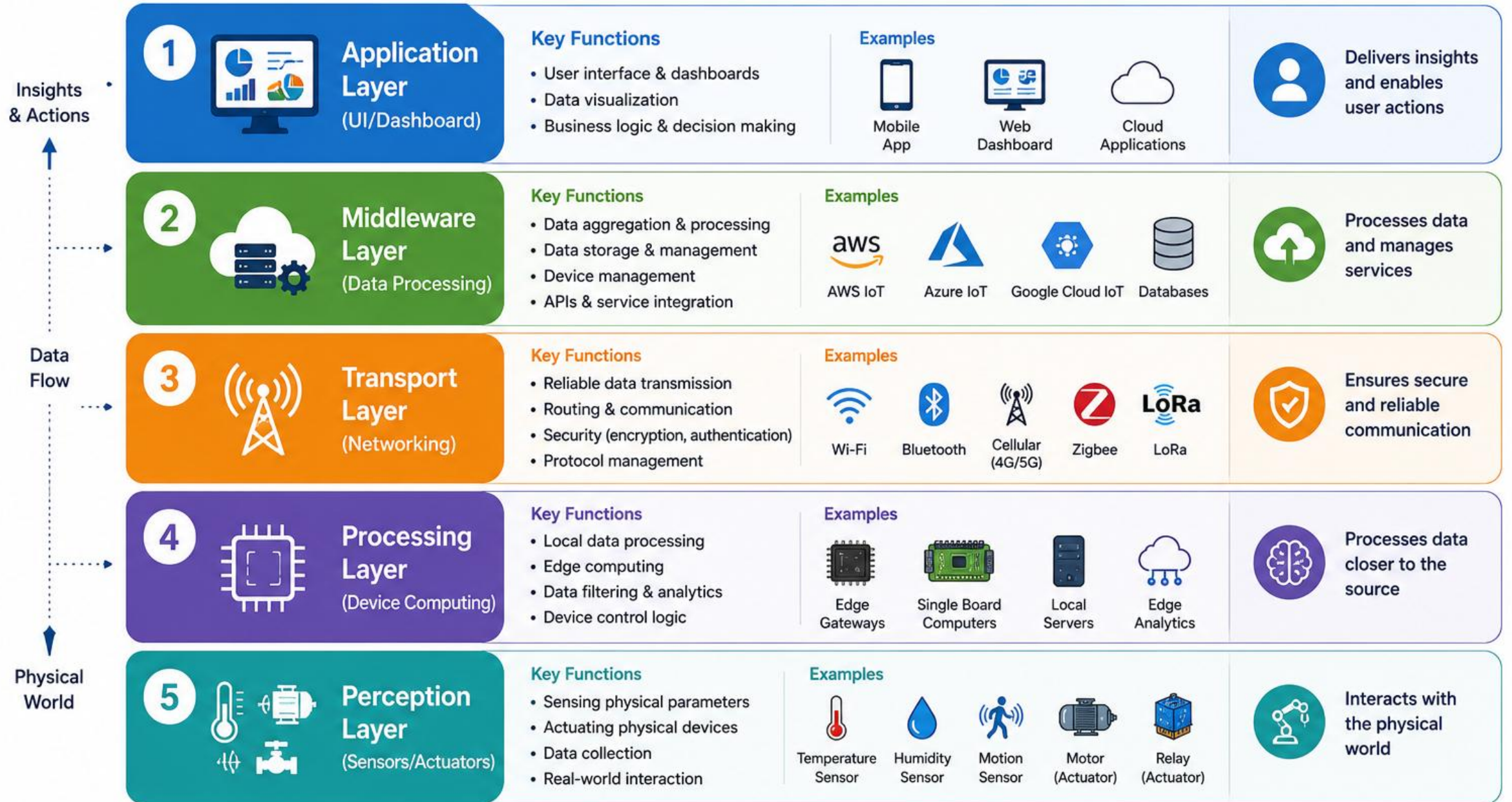
Things in IoT



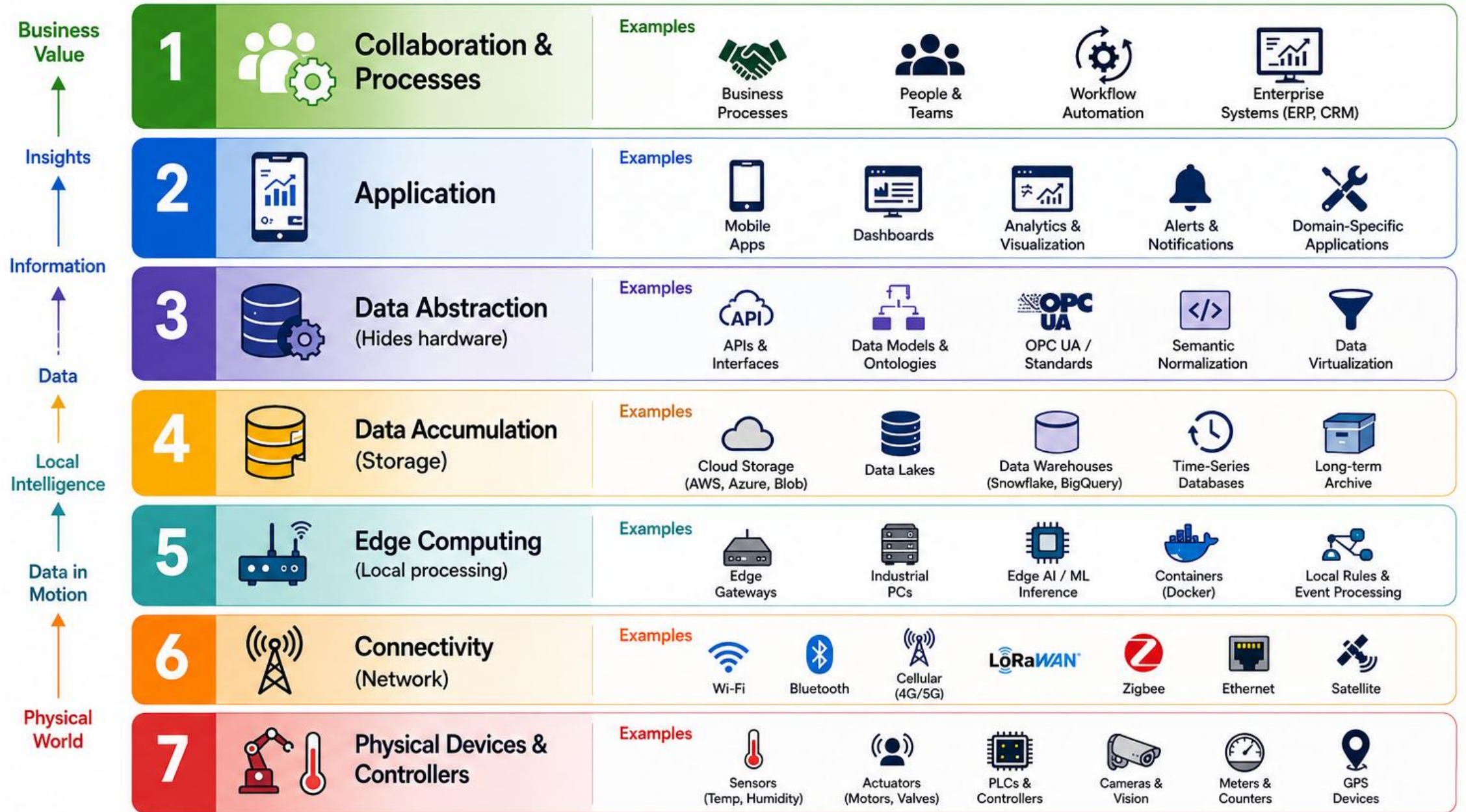
Three-Layer IoT Architecture



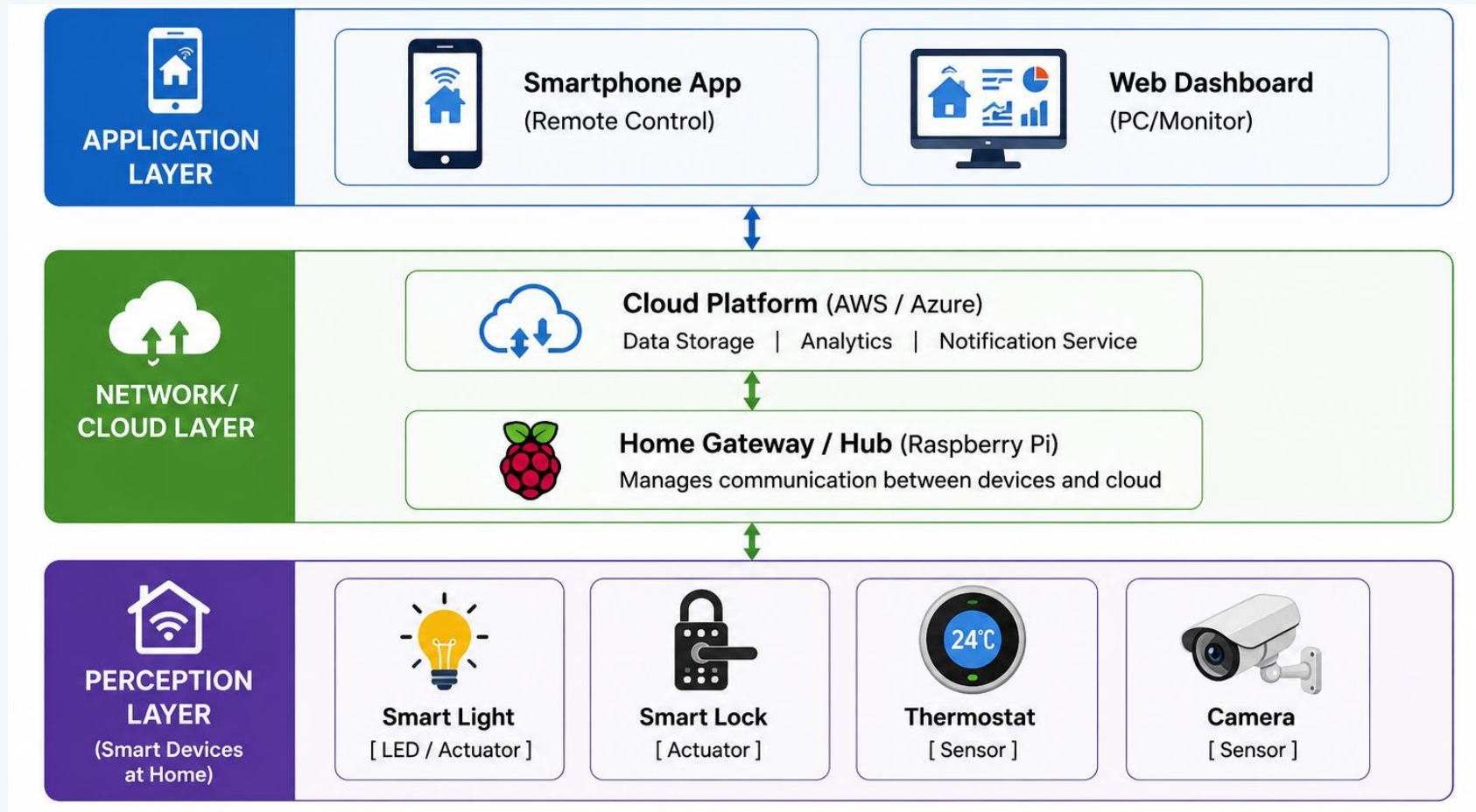
Expanded Architecture - 5 Layers



IIoTWF Architecture - 7 Layers



Architecture Example: Smart Home



IoT Communication Protocols

- Protocol Categories

- Application Layer MQTT, Coap, HTTP, DDS, XMPP
- Presentation Layer TLS/SSL, CBOR, JSON, XML
- Session Layer MQTT Sessions, CoAP Observe, WebSocket
- Transport Layer Tcp, Udp, Quic
- Network Layer Ipv4, Ipv6, 6lowpan
- Data link Layer IEEE 802.15.4, Wi-Fi, Ethernet, BLE, LoRa
- Physical Layer Radio Frequencies, Light, Electrical Signals

Detailed Breakdown of Lower Layers (1-3)

- **Layer 1: Physical Layer**

- Converts digital bits into physical signals (radio waves, light pulses, electrical)
- Defines frequency, modulation, and transmission power

Technology	Frequency Band	Range	Data Rate	Use Case
LoRa	Sub-GHz (868/915 MHz)	15+ km	0.3-50 kbps	Smart Agriculture, City
Zigbee	2.4 GHz	10-100 m	250 kbps	Smart Home, Industrial
BLE	2.4 GHz	10-50 m	1-2 Mbps	Wearables, Health
Wi-Fi	2.4/5 GHz	50-100 m	100+ Mbps	High-bandwidth IoT
NB-IoT	Licensed Cellular	10+ km	200 kbps	Smart Metering
LTE-M	Licensed Cellular	10+ km	1 Mbps	Mobile IoT

Detailed Breakdown of Lower Layers (1-3)

- **Layer 2: Data Link Layer**
 - Adds MAC (Media Access Control) addresses
 - Error detection and correction
 - Manages access to the physical medium
- IEEE 802.15.4 (Foundation for Low-Power IoT)
 - Physical Layer: 2.4 GHz, Sub-GHz
 - MAC Layer: CSMA-CA (Carrier Sense Multiple Access)
 - Uses 16-bit or 64-bit addresses
 - Supports mesh networking

Detailed Breakdown of Lower Layers (1-3)

- Wi-Fi (IEEE 802.11) CSMA/CA
 - Higher power consumption
 - Higher data rates
 - Direct internet connectivity
- Ethernet (IEEE 802.3) CSNA/CD
 - Wired connection
 - Reliable, high-speed
 - Industrial IoT (IIoT)

Detailed Breakdown of Lower Layers (1-3)

- **Layer 3: Network Layer**
 - Routing packets between networks
 - Addressing (logical addresses - IP addresses)

Protocol	Description	IoT Relevance
IPv4	32-bit addresses (4.3 billion)	Limited, mostly existing internet
IPv6	128-bit addresses (~unlimited)	Essential for IoT (billions of devices)
6LoWPAN	IPv6 over low-power wireless	Enables IPv6 on tiny devices

IoT Communication Protocols

- Common Protocols:
- MQTT Lightweight, publish-subscribe (Smart Home)
- CoAP Constrained devices, RESTful (Sensor nodes)
- HTTP Standard web protocol (Heavy, but universal)
- AMQP Secure, reliable messaging (Industrial)
- DDS Real-time, high performance (Industrial)

Smart Agriculture - IoT Protocol Stack

Layer	Technology	Function	Justification
Application	MQTT	Send data to farmer's app	Lightweight publish/subscribe
Presentation	TLS	Encrypt + compress data	Security + 50% smaller payload
Session	MQTT Sessions	Maintain offline message queue	Handles intermittent connectivity
Transport	TCP	Reliable data delivery	Guarantees no data loss
Network	IPv6 + 6LoWPAN	Unique addressing + compression	Billions of devices, saves power
Data Link	LoRaWAN	Long-range transmission	15+ km range
Physical	Sub-GHz (868 MHz)	Radio signal transmission	Penetrates foliage, low power

Resource Management in IoT

- Most IoT devices have LIMITED:
 - Battery/Power
 - Processing Power (CPU)
 - Memory (RAM/Storage)
 - Network Bandwidth

Key Resource Management Techniques

- Energy-Aware Allocation
 - Edge Computing
 - AI/ML-based Management
 - Duty Cycling
- Battery-Powered Weather Sensor
 - Wakes up every 30 minutes
 - Measures temperature & humidity
 - Sends data via low-power LoRa
 - Goes back to sleep → Battery lasts for years!